Toward Multi-Recognizer Multi-Adversary Training for Robust Face Recognition

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***Abstract*—Manipulated face imagery poses a growing threat to biometric verification systems because modern face synthesis and identity-targeted attacks can alter similarity relationships in embedding space while remaining visually plausible. This project initially focused on building a practical local framework for running diverse face-manipulation attack families and evaluating them across multiple recognizers. The work has since been refined toward a stronger research question: how adversarial ensemble training can be extended more thoroughly and more appropriately to face recognition by explicitly accounting for adversary-family diversity, recognizer-family diversity, and embedding-space verification behavior. Importantly, this pivot does not discard prior work; the implemented attack backends, recognizer integrations, and evaluation infrastructure now form the basis of the new training-oriented study. The ArcFace training model is already integrated with the intended ResNet-100 backbone confirmed, while the final end-to-end training pipeline and exact training method remain under construction. The final paper will study whether multi-recognizer multi-adversary training improves robustness to seen and unseen manipulations while keeping clean verification performance within an acceptable range.**

***Index Terms*—face recognition, adversarial robustness, biometric security, face manipulation, ArcFace**

I. Introduction

Face recognition systems are increasingly deployed in security-sensitive settings such as identity verification, access control, and digital forensics. At the same time, face swapping, morphing, reenactment, and adversarial identity attacks have become more capable and accessible. This creates a practical security problem: a manipulated probe may remain convincing enough to mislead a recognizer even when the image is not genuine.

An initial response is to benchmark attacks and measure how often they succeed against existing recognizers. While this remains necessary, it is not by itself the strongest research contribution for the current project. A more compelling direction is recognizer-side hardening: instead of only measuring vulnerability, the recognizer itself should be trained or fine-tuned to become more robust to diverse manipulation families.

This midterm report summarizes the project's transition toward *multi-recognizer multi-adversary training* for robust facial embedding learning. The central idea is that robustness may depend not only on exposure to a strong single attacker, but also on a structured ensemble on both sides: multiple recognizers and multiple attack mechanisms with distinct roles. This report therefore focuses on completed groundwork, the rationale for the pivot, and the current plan for the final paper.

II. Background and Partial Related Work

Modern face recognition systems commonly rely on embedding-based learning, where facial images are mapped into a feature space designed for verification and identification. Margin-based methods such as ArcFace are strong baselines because they improve embedding discrimination under face-recognition objectives [1]. Related recognizer families such as CosFace and CurricularFace motivate the idea that robustness should not be studied through only one recognizer design.

Prior work has also shown that face-recognition systems are vulnerable to diverse manipulation families, including face swapping, morphing, and identity-targeted generative attacks [2]–[4]. These attack families differ in how they alter the input and how well they transfer across recognizers. As a result, a recognizer hardened against only one manipulation family may still remain fragile under unseen or cross-family conditions.

This motivates the refined project direction. Rather than treating attacks only as benchmark inputs, the final paper will investigate whether diverse adversary exposure during recognizer training produces stronger and more transferable robustness than strongest-single-attack training alone. At this stage, the literature review is still incomplete, but the conceptual basis is already clear: face-recognition robustness should be studied in an embedding-centered setting with both clean-performance and robustness tradeoffs reported explicitly [5].

III. Work Completed to Date

Substantial preparatory work has already been completed, and this progress remains directly useful despite the pivot in paper direction. First, multiple face-manipulation and attack-family implementations have been integrated into the local project environment, with practical smoke-test or verification paths established for a core subset. This means the project already has a usable adversary pool rather than only a conceptual training plan. The implementation notes also distinguish benchmark-faithful, approximation-grade, and smoke-test-only paths so later experiments can be reported honestly.

Second, recognizer and evaluation infrastructure has already been prepared. Multiple recognizers have been set up as part of the local framework, including modern and legacy references used for cross-recognizer evaluation. This matters because the revised paper direction depends on recognizer-specific and cross-recognizer analysis rather than a single pooled verification result.

Third, the ArcFace training model has already been brought into the project and the main training backbone has been fixed to ResNet-100. An earlier training check helped confirm that recognizer training in the local environment is practical, even though the final end-to-end training pipeline and final method details are not yet complete. This is still meaningful progress because the main implementation risk has shifted from backbone availability to pipeline assembly, attack scheduling, and evaluation design.

Taken together, these completed steps show that the project is not restarting. The current state already includes attack-family implementations, confirmed recognizer families, evaluation scaffolding, and a usable ArcFace training entry point. These now serve as the operational base for the final paper.

IV. Pivot in Research Direction

The project originally leaned more heavily toward attack benchmarking and broader evaluation infrastructure for manipulated-face security analysis. That work was still useful because it forced the construction of a local framework capable of running multiple attack families and multiple recognizers under practical device constraints. However, as the project matured, it became clear that a pure benchmark-oriented story would be less focused and less compelling than a training-oriented contribution centered on recognizer robustness.

The revised direction is therefore a refinement rather than a reset. The project now aims to study multi-recognizer multi-adversary training for robust face recognition. In this framing, previously integrated attack methods become the adversary pool used for robust training and evaluation, while the existing recognizer setup becomes the basis for same-family and cross-family robustness analysis. The preliminary ArcFace training test further supports the feasibility of this pivot by showing that recognizer fine-tuning can already be exercised in practice.

This change also sharpens the scientific question. Instead of asking only which attacks succeed against recognizers, the project now asks which training assembly best improves robustness: strongest single-attack training, primary-attacker-set training, or primary-plus-surrogate attacker training across recognizer families.

V. Current Method Direction

The current planned direction is to train or fine-tune a face recognizer against a structured adversary pool and evaluate whether robustness generalizes beyond the attacks seen during training. On the recognizer side, the ensemble is intentionally restricted to ArcFace, CosFace, and CurricularFace. ArcFace is already the main recipe with a confirmed ResNet-100 backbone, while CosFace and CurricularFace provide closely related but still distinct margin-based formulations for same-family robustness comparison.

On the adversary side, the plan is organized into a primary attacker set and a surrogate attacker set. The primary attacker set is PGD, BPFA, and DFANet, chosen to cover white-box optimization, modern transfer pressure, and face-recognition-specific feature-level attack pressure. The surrogate attacker set is AdvFaceGAN, Adv-Makeup, and Greedy-DiM, chosen to cover unrestricted dual-identity pressure, semantic makeup manipulation, and morphing pressure. This structure is intended to keep the attack ensemble grounded and role-based rather than treating every available attack as one flat list.

At this stage, the final end-to-end training pipeline, exact attack sampling policy, and final loss formulation are not finished yet. The final paper will therefore compare conditions such as no adversarial training, strongest single attack, primary-attacker-set training, and primary-plus-surrogate mixtures only after the pipeline is fully assembled and validated.

VI. Next Steps

The next project stage is to convert the completed infrastructure and confirmed model choices into a full experimental pipeline. Since ArcFace with ResNet-100 is already fixed and the attack-family scope is already defined, the immediate tasks are now to assemble the final training loop, finalize the attack sampling and scheduling policy, and lock the first comparison conditions for training: clean baseline, strongest single attacker, primary-attacker-set training, and primary-plus-surrogate training.

After that, the project will execute reduced local smoke experiments to validate data flow and training logic, followed by larger-scale training runs under the final configuration where available. The literature review and experimental sections will also be expanded substantially for the final paper.

References

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